
PS Analysis in Data Science: Problem Set 5

Problem 1. Let $f : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ be defined by $f(x, y) = \|x\|^2 - \|y\|^2$ for $x, y \in \mathbb{R}^n$ and $\|\cdot\|$ the standard norm in $\mathbb{R}^n$.

- (a) Determine all local extrema and all saddle points.
- (b) Visualize f and ∇f in Julia for $n = 1$.

Problem 2. Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. We consider the loss function

$$L : \mathbb{R}^n \rightarrow \mathbb{R}, \quad x \mapsto \|Ax - b\|^2.$$

- (a) Compute the Fréchet derivative of L .
- (b) Show that $x \in \mathbb{R}^n$ is a global minimum if and only if

$$A^\top Ax = A^\top b.$$

- (c) Implement the (affine) local approximation of L in Julia.

Problem 3. Consider the functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}, \quad g(u, v) = u - v.$$

Compute $(g \circ f)'(x, y)$.

Problem 4. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$f(x, y) = \begin{pmatrix} xy \\ x^2 + y^3 \end{pmatrix}$$

- (a) Compute the Jacobi matrix $J_f(x, y)$.
- (b) Evaluate $J_f(1, 1)$.
- (c) Determine the (affine) linear approximation of f near the point $(1, 1)$.
- (d) Use this approximation to estimate $f(1.01, 0.98)$.